

ASSESSMENT-01

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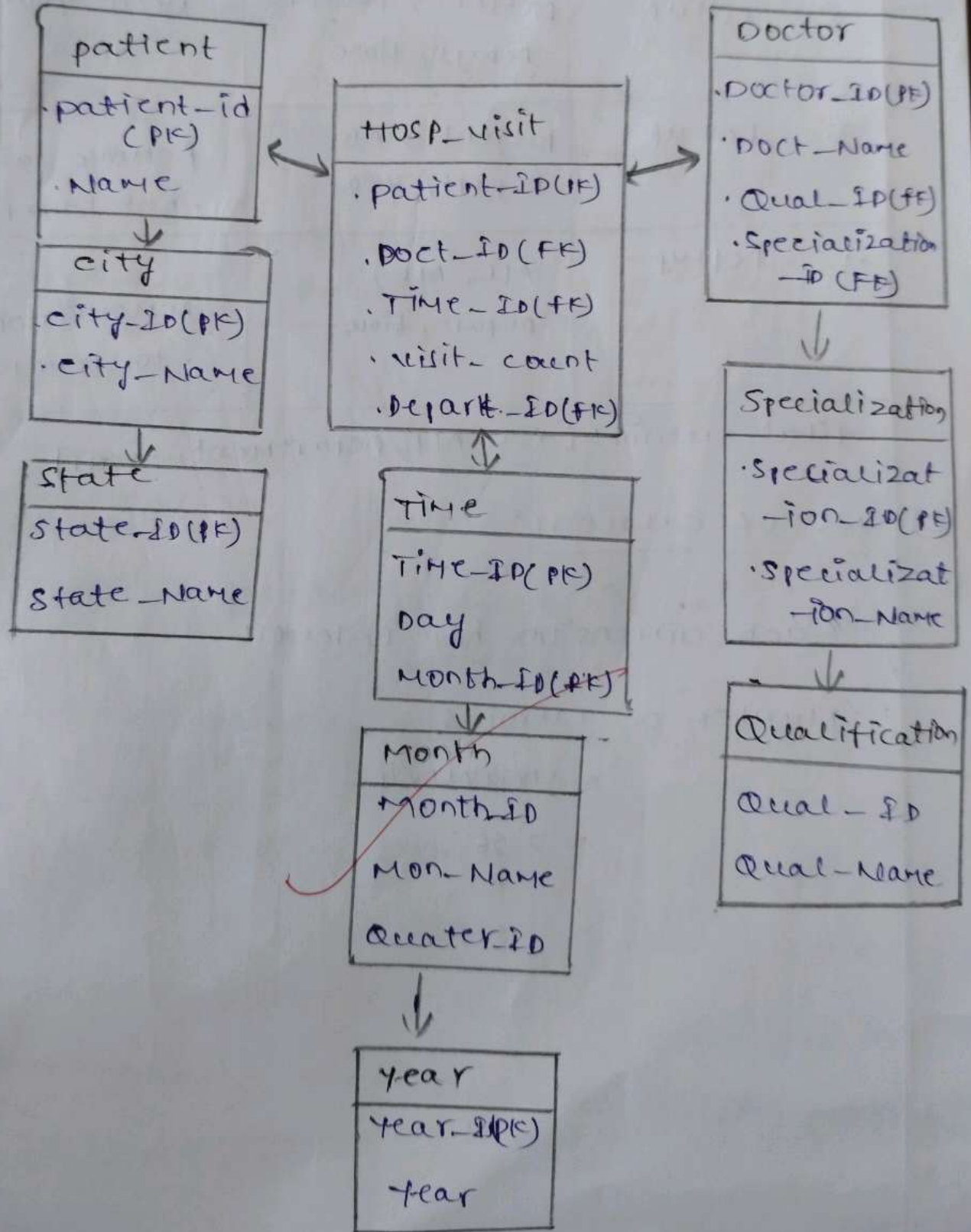
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SHORT ANSWERS

CSA1613

① City Hospital Datawarehouse :-

a) Snowflake Schema :-



⑥ OLAP Operations :

Step	operation	Resulting cuboid	Explanation
1	Start	patient, doctor, department	Base cuboid
2	Roll up	patient, doctor, depart, time	Need yearly data
3	Roll up	patient $\rightarrow$ All depart, year	Remove patient level
4	Roll up	ALL, ALL, depart, time	Remove doctor level

final cuboid: [ALL, ALL, department, year]

c) NO: of cuboids:

- Each dimension has 4 levels

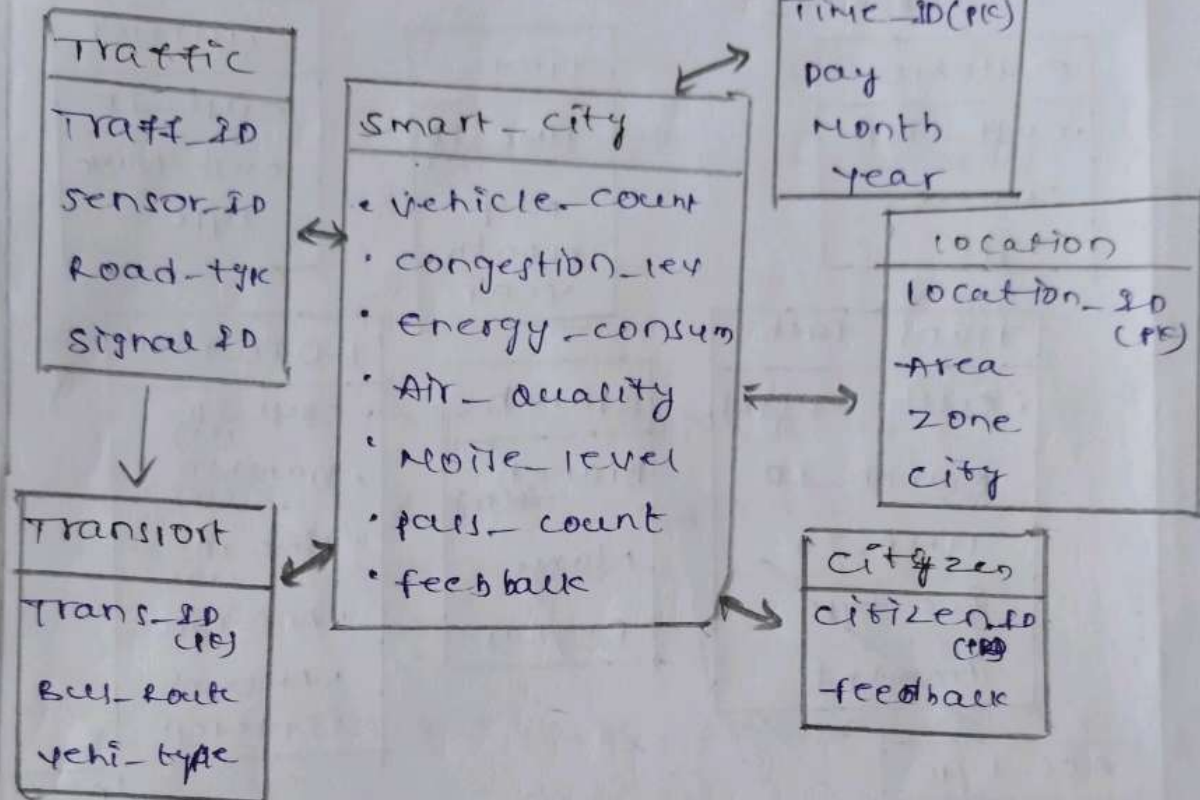
Number of cuboids

$$= 4 \times 4 \times 4 \times 4$$

$$= 256$$

Smart city data warehouse:-

Star Schema:

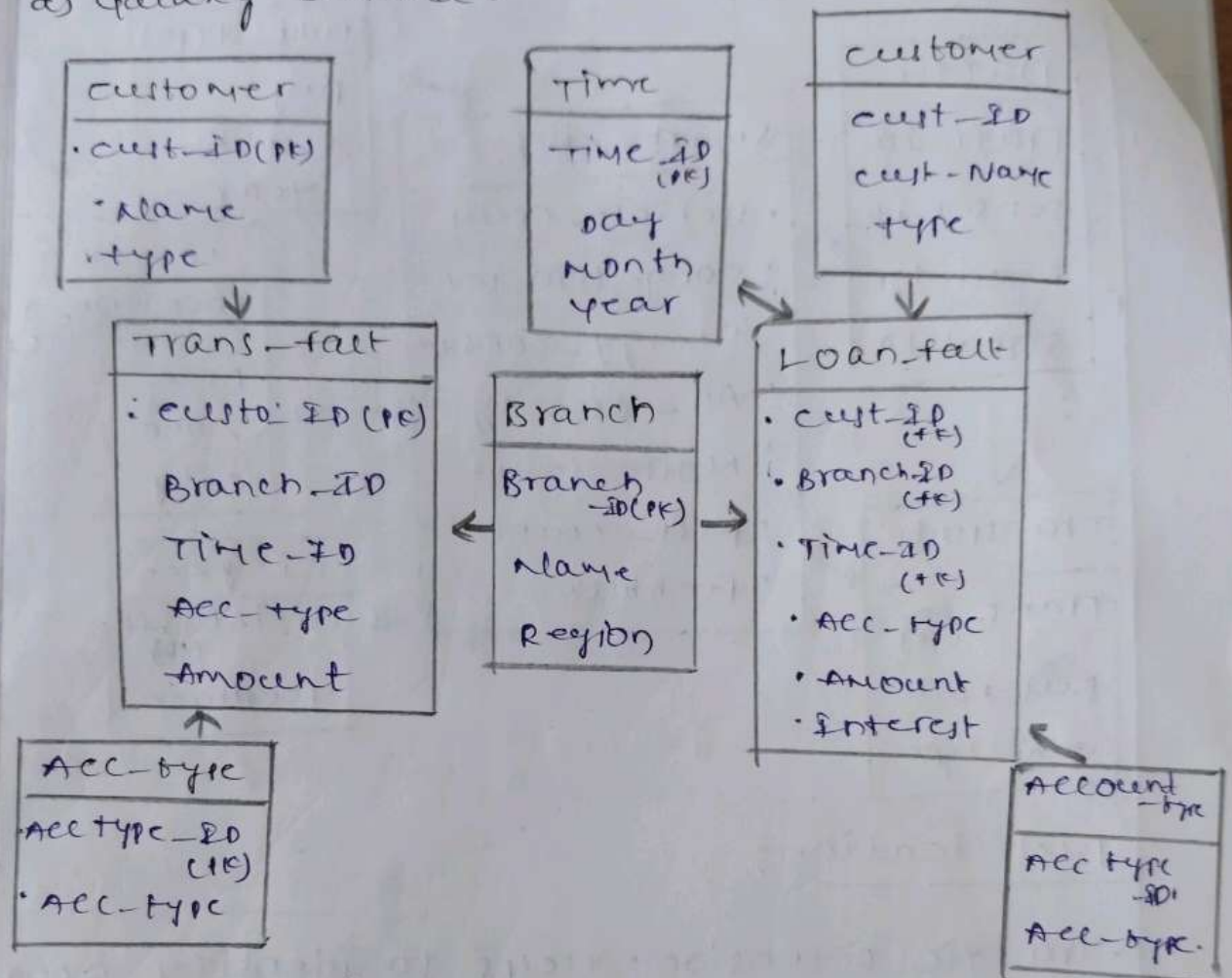


Justification:

- **Traffic dimension:** helps to identify congestion patterns.
- **location dimension:** compress data across different zones.
- **Time dimension:** enables analysis of daily, monthly, yearly trends.
- **Measures:** All are numeric and help in performance analysis and decision making.

③ Banking data warehouse :-

a) Galaxy schema :-



⑥ OLAP Operations :

Step	Operation	Resulting cubeoid
1	start	[customer, Branch, time, acc-type]
2	cust → ALL	[ALL, Branch, time, acc-type]
3	acc-type → ALL	[ALL, Branch, time, ALL]
4	time → year	[ALL, Branch, year, ALL]

No. of Cuboids :

- each level has dimensions
- for one cube $= 4 \times 4 \times 4 \times 4 = 256$
- for two $= 256 + 256 = 512$ cuboids.

(4) Manufacturing Company :-
Dimensions :-

- product : category, type, Batch.
- factory : Location, Dept, Line Number.
- Time : year, quarter, month, week
- supplier : Region, Rating, contract type.

Measure : Defect rate, production units, supplier, probability.

Operation	How to perform	Result
• Roll-up	• factory $\rightarrow$ ALL • product: Batch $\rightarrow$ type $\rightarrow$ category	• Avg defect rate by product category.
• Drill-down	• select a specific factory. • factory : location department	• Detailed defect rate by department and product type.
• Dice	• select product categories. • select factory locations.	• production units for high rated.

⑤ Retail chain :

Dimensions :

- product: category, sub category, sku
- store : Region, city, store id
- Time : year, quarter, month, day
- customer: membership, tier, Age, Gender.
- Measure : Inventory Turnover, total sales revenue, units sold

Operation	How to perform	Result
• Roll up	• store → ALL • product: sku → subcategory.	• Inventory turn-over by product category.
• Drill-down	• Select electronics in specific region • product: category → sku	• Sales Revenue by sub-category and sku region.
• slice 	• Customer: Gender = male • Membership tier = Bronze • Measure: Sales Revenue.	• Sales revenue for Bronzemale customers across all stores.